Stall Warning System – General Data

1. General

The aircraft stall warning system is triggered from the lift detector vanes located in the leading edge of the left wing. An active stall warning will trigger the Apex Stall Monitor and the independent Stall light located on the main instrument panel just below the ESIS. The Stall Monitor will trigger an Aural “Stall” annunciation, and the annunciation of Stall on the pilot's and co-pilot's PFDs.

Each lift detector has an integral anti-condensation heater operated by the pitot heater switch. Power to operate the system is from the 28 Volt left DC bus, routed through a 7.5 ampere PITOT HEAT L circuit breaker on the circuit breaker panel.

2. Adjustment/Test

A. Adjust Lift Detecting Vanes

Refer to Figure 1 for details.

Note The positions of the lift detecting vanes on the mounting plate assembly determine the airspeed at which stall warning is given. Therefore, whenever a vane is installed, an air test is required to determine the correct position of the vane on the mounting plate. The lower vane determines the airspeed at which warning is given in the range of 0° to 12° flaps down, and the upper vane determines the airspeed at which warning is given in the range of 13° to 40° flaps down. The data in Figure 1 gives the amount of change in airspeed produced by 1/8-inch movement of vane position for various flap settings.

1) Remove six screws securing lift detector vane mounting plate assembly to wing leading edge.

2) Carefully withdraw mounting plate assembly away from leading edge sufficiently to gain access to rear of mounting plate.

3) Using suitable marking instrument, mark rear of mounting plate and flanges of vane to be moved to show original installation position.

4) Using data given in Figure 1, change position of appropriate vane in direction required to produce warning at specified air speed.

Note The specified airspeed at which stall warning is given (STALL light illuminated) is 4 to 9 knots above stalling speeds. For stalling speeds at specified aircraft gross weight with various flap settings, refer to Flight Manual PSM 1-64-1A.

5) Secure detector vane mounting hardware and reinstall mounting plate to wing leading edge with six screws removed in Para 2.A.1).

B. Adjust Upper Vane Circuit Microswitch

Note Before attempting the following adjustments, ensure that the wing flaps are correctly rigged.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Using handpump, pump wing flaps to fully up position.

3) Set BUS TIE switch to NORMAL and ensure that CAUTION LTS NORMAL circuit breaker is energized. Operate CAUTION LIGHTS DIM switch and check that indication of a STALL is provided by Apex and the single square inverse video annunciation above the fuel selector knob, and is accompanied by a persistent aural warning.

4) Pump flaps to 12° down as indicated by flap position indicator.

5) Release cabin roof upholstery sufficiently to gain access to upper vane circuit microswitch positioned adjacent to flap actuator.

CAUTION **ENSURE THAT PITOT HEATER SWITCH IS OFF BEFORE TOUCHING STALL WARNING VANE, OTHERWISE A BURN COULD RESULT.**

6) With second operator actuating upper vane, loosen two switch adjusting screws and adjust switch until indication of STALL is provided by the display and the warning is heard, then back off until STALL indication and accompanied aural warning ceases. Tighten adjusting screws at this point.

7) Return flaps to fully up position and with upper vane actuated, lower flaps smoothly and slowly until STALL indication is displayed and accompanied by the aural warning, then stop pumping.

8) Check that flap position indicator reads 12° ± 2°.

9) If STALL indication is not displayed properly, nor is there the accompanied aural warning within range specified in Para 2.B.8), repeat adjustment procedure as necessary until STALL is displayed properly, (refer to Para 1.) and is accompanied by the persistent aural warning.

10) Replace upholstery released in Para 2.B.5) and disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

C. Function Test Stall Warning System

1) Ensure that wing flap system is operating properly and that indicating system is accurate.

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Using handpump, pump wing flaps to fully up position.

4) Set BUS TIE switch to NORMAL and ensure CAUTION LTS NORMAL circuit breaker is energized. Operate CAUTION LIGHTS TEST switch and check that STALL is provided by the display, as indicated in Para 1.

Note Two people are required to do Para 2.C.5) through Para 2.C.9).

CAUTION **ENSURE THAT PITOT HEATER SWITCH IS OFF BEFORE TOUCHING STALL WARNING VANE, OTHERWISE A BURN COULD RESULT.**

5) Actuate lower vane gently to complete circuit and hold vane in that position. Pump flaps from fully up to fully down and check that STALL is provided by the displays as indicated and the audible “stall” stall warning is heard through the headset.

6) Release lower vane and return flaps to fully up position.

7) Actuate upper vane gently and check that there is no STALL indication displayed and no warning heard.

8) With upper vane actuated, pump flaps slowly and smoothly down until STALL indication is displayed and accompanied by the aural warning, then stop pumping. Check that flap position indicator reads 12° ± 2°.

9) With upper vane actuated, continue pumping flaps to fully down. Check that STALL indication is displayed and the aural warning heard.

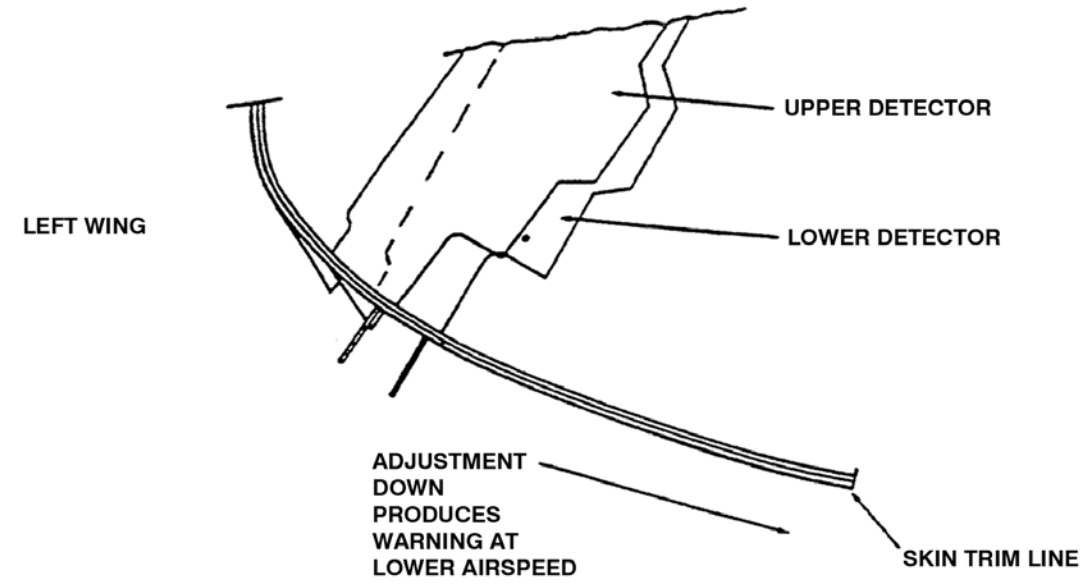
10) If light does not illuminate within specified range, adjust upper vane microswitch in accordance with procedure given in Para 2.B.

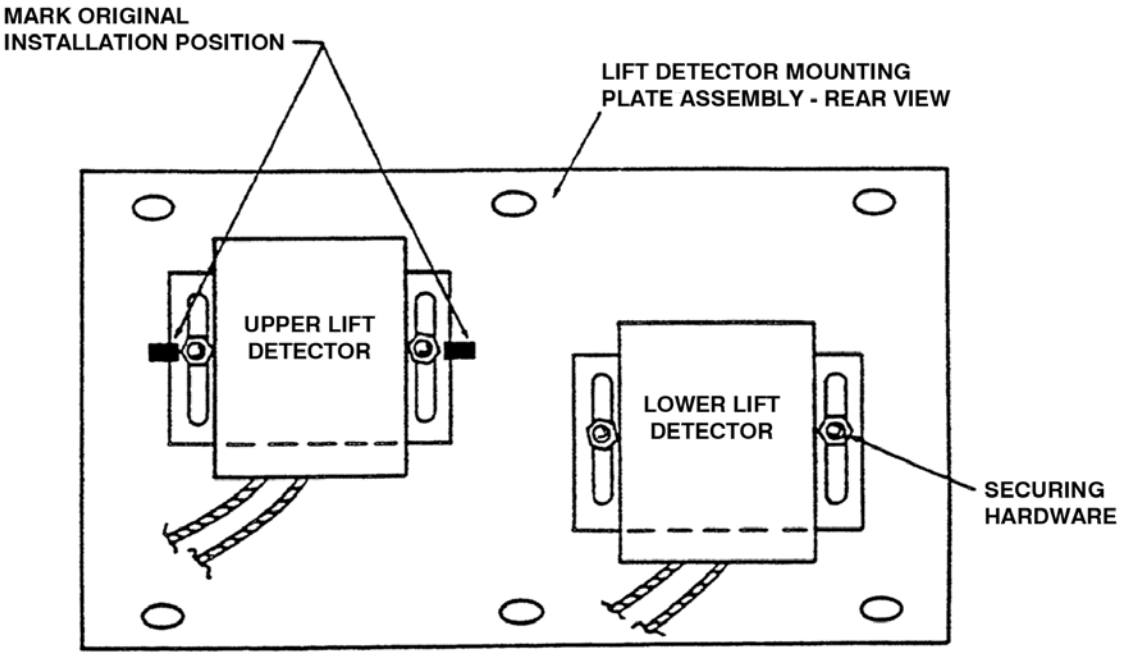
11) When the conditions of Para 2.C.10) are satisfied, select PITOT HEAT switch to ON and check that the stall warning vane heaters warm up.

CAUTION **DO NOT LEAVE PITOT HEATER SWITCHED ON FOR MORE THAN ONE MINUTE.**

12) Select PITOT HEAT switch to OFF.

13) Disconnect DC power source from aircraft (refer to 12-00-00, Servicing – General Data, Para 2.).





LIFT DETECTOR	FLAP POSITION (DEGREES DOWN)	CHANGE PRODUCED BY 1/8-INCH MOVEMENT OF LIFT DETECTOR
		KNOTS
LOWER	0	1.87
LOWER	10	1.19
UPPER	20	0.81
UPPER	40	0.47

Calibration Data for Lift Detector Vane
Figure 1